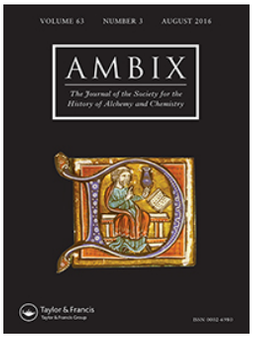


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# Orchil and Tyrian Purple: Two Centuries of Bedfords from Leeds

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A two-century-old archive relating to a Leeds dye manufacturer emerged a number of years ago from a Devon attic. Most items in the archive date from the mid-nineteenth century to the first quarter of the twentieth, a period of dye history when natural dyes overlapped with and were gradually replaced by synthetics. The archive contains material relevant to three generations of the Bedford family's manufacturing and research successes, as well as its close connections to the family of William Henry Perkin. A major portion of the archive's contents is connected to the trade in orchil, a purple-producing dye from a lichen on which the early fortunes of the company were based. A small envelope signed by Charles Samuel Bedford states that it contains "Tyrian Purple." After a historical and chemical investigation, we found that this dyestuff was indeed from mollusca, confirming that this vat-dyed textile sample is the most modern historic sample of Tyrian Purple yet found.

## Introduction: The Bedford archive

Almost a decade ago, friends of the first author asked her to look at an archive they had been keeping in their attic in Hatherleigh, a town in West Devon. They were about to move to a smaller house and the collection's next destination was to be a Devon barn. The contents of the archive had belonged to three generations of the Bedfords, a prominent family of industrial chemists based in Kirkstall Road, Leeds.

The Bedford family included three dye chemists successively named James Bedford, whom, to avoid confusion, we will henceforth refer to as Bedford I, Bedford II, and Bedford III. Mr Charles Chalcraft, the current owner of the archive, is the grandson of Charles Samuel Bedford, of the third Bedford generation. According to Chalcraft, the physical archive was assembled by James Bedford III

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FIGURE 1 Shipping invoices and ledgers from the Bedford archive.

between the time of his retirement from the company in 1914 (when he became Lord Mayor of Leeds) and his death in 1925. Since Bedford and his wife had no children, the archive passed to his younger brother, Charles Samuel Bedford, and through Charles's daughter, Jill Bedford, to her son, Mr Chalcraft.[1]

The Bedfords' family history, their company, and their role in the dyeing industry are revealed by the letters, diaries, and material objects found in this collection. Over the course of several months, the contents of the archive, housed in a collection of boxes and chests, were sorted, examined, and relocated for cataloguing and eventual public study.[2] Most of these date to before 1930, and include items related to trade, administration, chemistry, the purchase of land, buildings, patents, and inventories. Among the wide variety of objects in the archive are many sample books, ledgers, publicity materials, stockbooks, and exhibition catalogues (Figure 1). The boxes contained a number of powdered chemicals and dyes in jars; some, unfortunately, were broken and surrounding papers contaminated or destroyed (Figure 2). The collection also includes botanical specimens of orchil-producing lichens from the Canary Islands, a few indigo seeds, and some dried dyestuffs, such as saffron (Figure 3). Another significant item is a "Family History" compiled by Bedford III in the form of a large, handwritten album that contains letters, newspaper cuttings,

[1] Conversations with Charles Chalcraft commenced in 2007 and continue to the present day.

[2] Regarding the physical collection, there was often no discernible system in the way things were stored in the boxes and trunks. It is possible that Jill Bedford repacked the material when she unsuccessfully tried to find a home for the collection in the 1960s. The archive is currently being catalogued by Dr Howard Varley, late of Yorkshire Chemicals. Records additional to this archive owned by Charles Chalcraft, a fifth generation Bedford, have surfaced through a group of ex-Yorkshire Chemicals employees, their families and descendants. These items have been added to those being catalogued and will be curated by the West Yorkshire Archive Service (WYAS) once cataloguing is complete. Certain items, such as the Bedford Family History, will be retained by Mr Chalcraft, who is happy to discuss the archive with researchers in the future. He can be contacted via the first author.

FIGURE 2 Loose dye powder from broken glass containers.

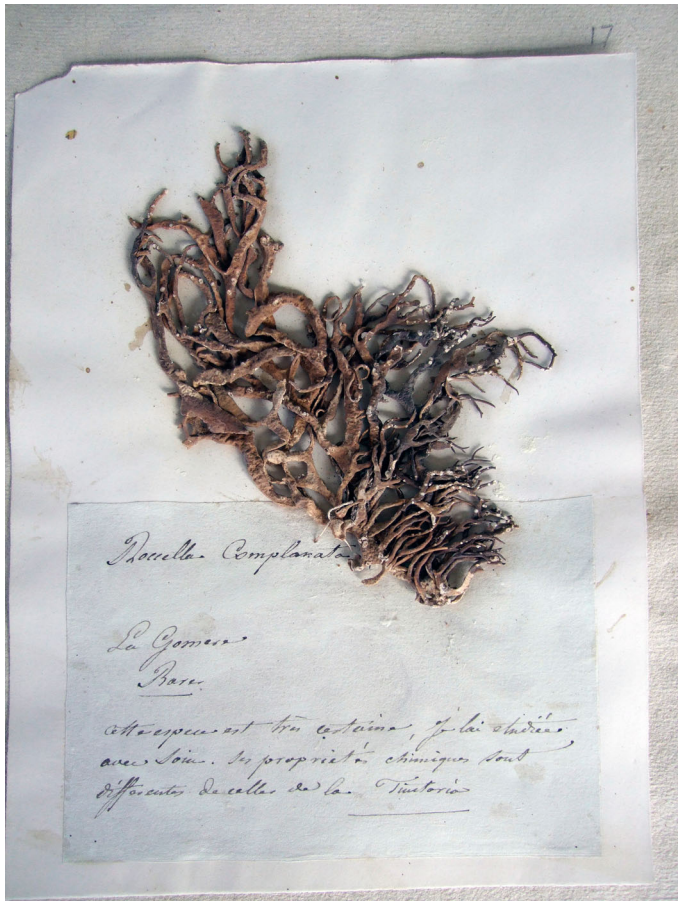


FIGURE 3 Specimen page for a Canarian dye lichen from the botanical collection.

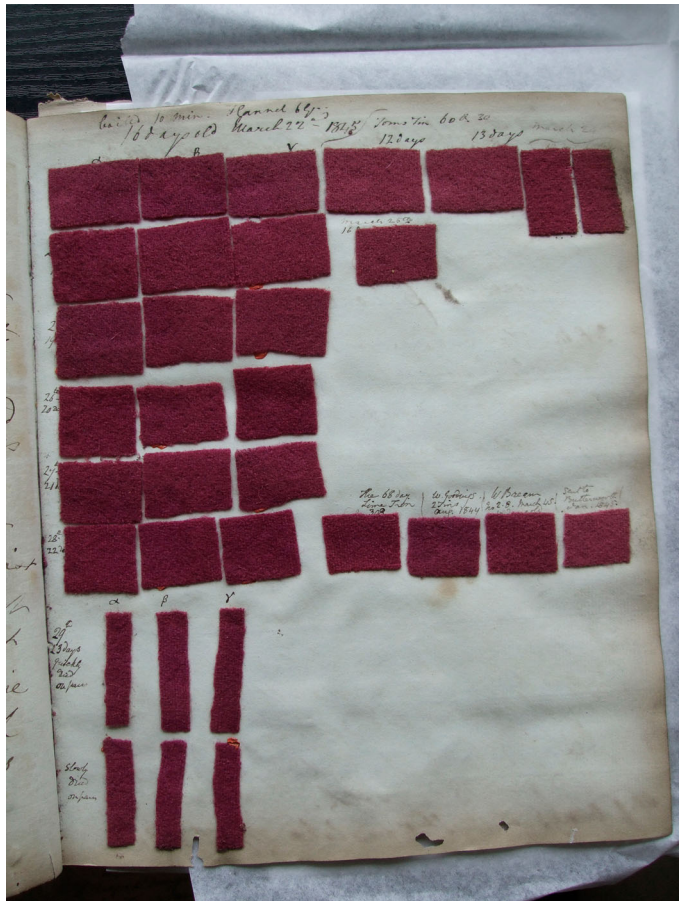

FIGURE 4 Page from a dyer's notebook from the mid-1800s containing purple woollen swatches dyed with orchil.

photographs, and other documents, in addition to a history of the family and the company's development. The result is an orderly, well-referenced, self-contained record which includes some key documents related to the development of the business.

The archive is a rare example of a largely continuous chain of documents and artefacts from the early 1800s through to the mid-twentieth century. During its 150 year occupancy of the Kirkstall Road premises, the family company changed its name several times, but it never underwent a takeover, a transition point at which records are often discarded. These records of an enterprising and industrious family offer us an opportunity to study an absorbing period of dye manufacture, chemistry, and trade. They span the decades when natural dyestuffs were gradually replaced by synthetics. They also reveal that the Bedfords' early success as dye manufacturers rested on the successful processing and marketing of a single commodity: the lichen purple dye known as orchil (Figure 4), and the related dye known as

cudbear. For instance, the archive includes the notebook of F. W. Marshall, a dyer whose family's company was later merged with that of the Bedfords.[3] Marshall's notes, written around 1845, record stages of processing orchil in written observations and dyed wool flannel samples, noting the stages at which the dye was at its most effective. Other notebooks contain detailed records of the purchase and processing of orchil lichen, as well as indigo and tannins which the company processed in the later part of the nineteenth century. Also in the archive is an envelope containing several samples of silks dyed in the family's patented "magogs" (mechanised orchil vats), using orchil lichen that was probably gathered on a holiday tour to Norway, a testament to the family's ingenuity and enterprise in the processing of orchil. However, orchil is not the only purple dye represented in the archive: a small envelope, found caught in the hinge of one of the trunks, contains a dyed sample labelled "Tyrian purple."

In this paper, we will use the contents of the archive to trace how successive generations of the Bedford family, as well as their close connections in the Perkin family, used their chemical knowledge and skills to investigate commercial products. In particular, analysis of the "Tyrian purple" sample shows how the Bedfords continued to experiment with new substances, even when the orchil side of their business was well established.

## Historical orchil processing

Orchil, the "cash crop" of the Bedfords, appears in innumerable recipes from ancient times up to and including the nineteenth century, and was in demand as a standard dyestuff throughout Europe long before the synthetic dye age initiated by William Henry Perkin in 1856. Indeed, the use of orchil is documented from at least the Aristotelian Age. In the early third century BCE, Theophrastus wrote:

> In Crete there is an abundant and luxuriant growth on the rocks close to land, with which they dye not only their ribbons, but also wool and clothes. And as long as the dye is fresh, the colour is far more beautiful than the purple dye; it occurs on the north coast in greater abundance and fairer, as do sponges and other such things.[4]

In the first century CE, Pliny the Elder wrote about orchil using the Greek "phycos" (φύκος) for lichen:

[3] The Marshall family's company was one of those that in 1900 amalgamated with Wood & Bedford, mentioned below.

[4] Theophrastus, *Enquiry into Plants*, trans. A. Hort, vol. 1, bk IV, 6.5: 333, https://archive.org/stream/enquiryintoplanto1theouoft#page/333/mode/2up (accessed 25 January 2014). The Greek "phycos," from which the English word "fucus" is derived, is a general term for lichens. In fact, in the Index of Plants in this translation there are six different definitions of the word φύκος as a plant typically associated with the sea, e.g. seaweed, sponge, or alga (Theophrastus, *Enquiry*, 483).

> The phycos which grows on the rocks in the neighbourhood of Crete, is used also for dyeing purple; the best kind being that produced on the north side of the island, which is the case also with sponges of the very best quality.[5]

Also in the first century, Dioscorides mentioned lichens in a long list of herbal medicines, albeit only with regard to their medicinal properties, and not as dye plants.[6] Similarly, the Biblical and Talmudic colourants associated with the Hebraic word "pukh," which is probably etymologically equivalent to the Greek "phycos," do not refer to a textile dye, but rather either to a cosmetic or a medication for the eyes.[7]

Lichens are generally any of numerous complex plant-like organisms made up of fungi and an alga or a cyanobacterium growing in symbiotic association, typically on a solid surface such as a rock. Orchil (also called orcein) is a natural purple dyestuff produced from various lichens, primarily from *Roccella tinctoria*, and it consists of a number of related dye components whose chemical structures were first elucidated by Musso in the 1950s.[8] Throughout its entire history, manufacture of orchil was never a straightforward process, because the dyestuff is not immediately available from the lichen; a kind of fermentation must first take place to break down the colourless polyphenolic depsides and depsidones, which are the dye precursors.

Generations of authors have provided recipes for preparing and dyeing with orchil. The basic method involves crushing lichen in troughs or vats containing alkaline materials, such as stale urine, plant ashes, or lime. Regular introduction of air into the mixture is essential, particularly in the first stages, and this was effected by paddles, churns, or stirring rods. Over a period of days the mixture turns brown, and then purple. Keeping the mixture warm hastens the process, which can take from days to weeks.

Later recipes and authors reflect newer knowledge, such as geographical sources of the best lichen dyestuff.[9] In 1814 Edward Bancroft recorded making orchil with

[5] Pliny the Elder, *The Natural History*, trans. J. Bostock and H.T. Riley (London: Taylor and Francis, 1855), bk XIII, 48, http://www.perseus.tufts.edu/hopper/text?doc=Perseus%3Atext%3A1999.02.0137%3Abook%3D13%3Achapter%3D48 (accessed 25 January 2014).

[6] Dioscorides, *De Materia Medica*, trans. T. Osbaldeston (Johannesburg: Ibidis Press, 2000), Book IV: "Other Herbs and Roots," http://www.cancerlynx.com/BOOKFOURROOTS.PDF (accessed 25 January 2014); he mentions "Leichen" (4-53, on 595), "Bruon Thalassion" (4-99, on 651), and "Phukos Thalassion" (4-100, on 651).

[7] In the Hebrew Prophets (Kings II 9:30 and Jeremiah 4:30), there are two citations of the word *pukh*, referring to an eye cosmetic. The multi-volume Babylonian Talmud, written in Aramaic and Hebrew in the Roman and Early Byzantine Periods, also mentions the word *pukh* (in Tractate Shabbat 109a), but in the context of an eye medication, not cosmetic; see Zvi C. Koren, "Historico-Chemical Analysis of Plant Dyestuffs Used in Textiles from Ancient Israel," in *Archaeological Chemistry: Organic, Inorganic and Biochemical Analysis*, ed. Mary Virginia Orna (Washington, DC: American Chemical Society, 1996), 269–310. R. J. Forbes, *Studies in Ancient Technology* (Leiden: E. J. Brill, 1964), 109, mentions the Hebrew term *pukh* as a lichen used for eye makeup, but infers that this is the same *pukh* that is also the orchil dye, which it was not. Some authors (e.g. Annette Kok, "A Short History of the Orchil Dyes," *The Lichenologist* 3 [1996]: 248) have mistakenly cited Ezekiel 27:7 as referring to lichen dyes; however, the two sacral and royal dyeings that are mentioned, *tekhelet* and *argaman*, are well-known Hebraic terms for the two types of molluscan dyeings – blue-purple and red-purple, respectively; see Zvi C. Koren, "New Chemical Insights into the Ancient Molluskan Purple Dyeing Process," in *Archaeological Chemistry VIII*, ed. R. Armitage and J. Burton (Washington, DC: American Chemical Society, 2013), 43–67.

[8] H. Beecken et al., "Orcein and Litmus," *Biotechnic & Histochemistry* 78 (2003): 289–302.

[9] M. Hellot, M. Macquer, and M. le Pileur d'Apligny, *The Art of Dyeing Wool, Silk and Cotton* (London: Scott Greenwood & Co, 1901), 209; Giovanni Rebora, *Un Manuale di Tintoria del Quattrocento*, ed. Universitá degli Studi di Genova (Milan: Dott. A. Giuffrè, 1970), 97–98; Dominique Cardon, *Natural Dyes, Sources, Tradition, Technology*

distilled, as opposed to "raw," urine, noting the great improvement in colour due to elimination of the "volatile parts of solid human waste."[10] It is clear through these generations of recipes that orchil was a common dyestuff until the mid-nineteenth century, used on its own or in combination with other dyes. The trade in lichen dyestuff throughout these years was therefore substantial, and Europe-wide.

Orchil is a "substantive" dye, that is, it can be used without a mordant, which is especially advantageous when dyeing silk because mordants can make the fabric harsh to the touch. Used on its own, orchil will dye protein fibres (wool and silk) various shades of pink and purple. By contrast, it is not successful on cellulose fibres, although some progress was achieved in the nineteenth century in printing linen and cotton with the related dye cudbear.[11] This process produced a wide range of purple shades without the expense of a two-stage blue-red overdyeing. Overdyeing was typically accomplished via the blue indigo pigment – produced from woad or the indigo plant – and a red dye extracted from a madder plant, or from scale insect dyes, such as kermes and cochineal.

Though freshly dyed colour is unrivalled in a plant dyestuff, the colour is often short-lived because orchil has mediocre to poor fastness to light. It is surely the beauty and utility of the colour that explains its use in so many recipes, despite its fugitive nature. Perhaps more surprising, given technique, cost, and display, is the use of orchil reported in costly tapestries and textiles.[12] However, various techniques were applied to preserve the colour as long as possible.[13]

Cudbear and litmus are colourants related to orchil. Cudbear is a dried, powdered form of orchil, patented by the brothers George and Cuthbert Gordon in 1758 and manufactured first at Leith. The name "cudbear" is derived from "Cuthbert," their mother's maiden name.[14] Their success is thought, in part, to be due to efficient urine distillation, which produced a better colour. In later years, and with considerable secrecy, cudbear was produced at Dunchattan, near Glasgow, under the management of George Macintosh, until the Macintosh Company ceased production in 1852.[15] Litmus was made from the lichen *Ochrolechia tartarea*, and was produced

[9] *Continued*
*and Science* (London: Archetype Publications, 2007), 490–91; Gioanventura Rosetti, *The Plictho*, trans. S. Edelstein and H. Borghetty (Cambridge, MA: MIT Press, 1969), 120.

[10] Edward Bancroft, *Experimental Researches Concerning the Philosophy of Permanent Colours* (Philadelphia, PA: Thomas Dobson, 1813), vol. 1, 224.

[11] George Wagstaffe, *Official Catalogue of the Great Exhibition of the Works of Industry of All Nations, 1851*, ed. R. Ellis (London: Commissioners for the Exhibition, 1851), 34 (entries 68 and 72).

[12] Maria Jordan, "Recreating the Life of a Tapestry: Fading Dyes and the Impact on the Tapestry Image," *The Institute of Conservation, Textile Group Postprints* (2013): 43–57; Anna Maria De Strobel, "Weaving the Sistine Chapel Tapestries," in *Raphael: Cartoons and Tapestries for the Sistine Chapel*, ed. M. Evans et al (London: V&A Publishing, 2010), 33–37. As we have seen, orchil was used for "nuancing" and as a means to enrich colours. Clearly, there was some fraudulent abuse of this dyestuff, which would have been undetectable at point of purchase. Bancroft records an instance of restitution of several thousand pounds levied on a London dyer who had used cudbear instead of the indigo and cochineal he had claimed: Bancroft, *Experimental Researches*, vol. I, 225.

[13] Hellot et al. (n. 10) write of orchil-dyed silks: "Acids destroy them and they so quickly fade that to keep silks of this colour, if you would preserve them fresh, it is necessary to keep them as much as possible from the air."

[14] W. T. Johnston, *Cudbear Dye and its Discovery by Cuthbert Gordon (1730–1810)* (Livingston: Officina, 1995).

[15] Archibald and Nan Clow, *The Chemical Revolution: A Contribution to Social Technology* (London: Batchworth, 1952), 214.

as a dye in the Netherlands from the middle ages on, mixed with powdered chalk or gypsum and sold in small oblong cakes.[16] Of course, litmus paper was also long used as a colour reagent. In the twentieth century orchil was used as a food colourant (listed as E121), but was later banned from such use by the European Union;[17] more recently it has been used as a stain in microscopy.

Varieties of orchil lichens come from different species and can have very different appearances. They are generally found growing on rocks or on trees, and they can be crusty in appearance, like *Ochrolechia tartarea*, or more biscuit-like, as in *Lasallia pustulata*. Some orchil lichens which grow on trees (*Roccella spp*) can also be found on rocks, and may be branched and strap-like in appearance. The crusty or biscuit-like lichens, which were gathered by pulling or scraping them from the rock surface, were known in the nineteenth century trade as "moss" or "rock moss"; those such as the *Roccellae* were known as "weed."

Experience in manufacture would have taught punctilious dye makers that individual lichen types (and sources) might require different preparations and treatments, such as a variation in fermentation times. Extensive experimentation has shown that different varieties also produce different shades of varying quality, and "weeds" were more valuable than "mosses." For instance, *Roccellae* are often richer in dyestuff than "mosses" and normally require less cleaning. Rock-growing lichens were often deliberately contaminated by dirt and stones, which added to their weight.[18] Such knowledge was essential in the dyeing business. As we will see, the Bedfords and their connections actively experimented on lichens and other substances, both in the course of developing new commercial products and possibly also for their own personal interest.

## The Bedfords: James I (1795–1842), II (1824–1903), and III (1856–1925)

The Bedford family's first manufacture of orchil started with the work of a young chemist, James Bedford I, the son of the principal manager for a colliery railway at Middleton, outside Leeds. In 1810, at the age of fifteen, Bedford was apprenticed to a Mr Ellershaw as one of the archive's letters indicates, "to be taught, learned, and informed in the Art, Trade, and Mystery of a Chemist and Druggist" in Briggate, Leeds.[19] On completing his seven-year apprenticeship he went into equal partnership with a Mr Joseph Walker, also in Briggate, but the alliance was short-lived. Although Walker was a respected chemist and botanist, his love life had borne him disappointment and he turned to the addictive laudanum bottle – one

[16] Cardon, *Natural Dyes*, 492.
[17] https://www.food.gov.uk/science/additives/enumberlist (accessed 18 September 2016).
[18] Lázaro Sánchez-Pinto, "A Brief History of the Canary Weed," in *Flecheten Follmann. Contributions to Lichenology in Honour of Gerhard Follmann*, ed. F. J. A. Daniels et al. (Cologne: Botanical Institute, University of Cologne, 1995), 549.
[19] From the Bedford Family History (unpublished).

assumes off the company shelves. Not surprisingly, Bedford terminated their association in 1821 and started out in business on his own, partly in the refining of oils.

At some point, however, he began to make orchil and cudbear. In a letter written in December 1827, a Mr Taylor of Liverpool offers his experience and services in orchil-making to the thirty-two-year-old Bedford. He observes drily of the dyestuff: "Holmes & Son of this town have made a large fortune by making it and conduct the concern with much pretended mystery and secrecy."[20] An invoice relating to the purchase of "moss" and "weed" in February of the same year suggests that Bedford quickly began learning the practicalities, if not the pretended mystery. The address on this invoice shows that his work with orchil began with experiments at the Briggate premises. Two invoices from George Macintosh and Company, dated 1829, also record his purchases of orchil from Glasgow: purchases from a prominent supplier which may have been for comparison with his own product. By the time Bedford moved residence to Rockingham Street in around 1834, he was styling himself as a manufacturing chemist and "maker of cudbear and archill."[21] At some point thereafter he moved his works from Briggate to Kidacre Street.

By 1824 James Bedford I was father to James Bedford II, who attended Leeds Grammar School, and later, at the age of fifteen, Hull College, where he was a bright scholar, winning many prizes. Additionally, he was accustomed to helping his father with the business in his school holidays, experience which would later stand his family in good stead. When his father died unexpectedly in 1842, Bedford II, then aged eighteen, was recalled from Hull College to run the family business and support his mother and two younger sisters.

His late father's money was locked into the company, probably in stocks of lichen and the orchil already in preparation. Bedford I is recorded as having paid £300 per ton for lichen during some periods of manufacture.[22] Failure of these costly vats during fermentation caused major financial loss and immense anxiety.[23] One of the orchil manufacturer's greatest skills was to judge when the mixture was at maximum effectiveness and was ready to use. Already in the fifteenth century Giovanni Rebora had observed: "those who keep it for too long, it will evaporate from them and they will lose both their capital and their profit."[24] At the time of the death of Bedford I a large fortune may have been tied up in the company's orchil, with financial shortfalls to be rectified by a young man with relatively little experience.

Bedford II now had the ultimate responsibility of managing the family's orchil business. Confident management of the vats as well as efficient sourcing and control of orders and supplies would have been necessary to maintain family income and realise the invested capital. Profiting from orchil required skill,

[20] Bedford Family History.
[21] Archill is an old name for orchil, used on a visiting card kept in the Bedford Family History.
[22] Bedford Family History. Using the retail price index this is about £25,000 at 2016 values, calculated via Measuring Worth, http://www.measuringworth.com (accessed 18 March 2016).
[23] Bedford Family History.
[24] "Chi li tien tropo I va in fumo et perdense el cavedal e'l pro." Rebora, *Un Manuale di Tintoria*, 98 (see n 9).

because the process of dye preparation was not straightforward and raw materials were often expensive. Prices were frequently inflated because lichen, which cannot be cultivated and grows slowly, was essentially a non-renewable resource. Once an area was stripped out it rarely recovered, and when stocks in individual areas were depleted from over-harvesting traders had to move on through what became a succession of geographical sources. Prices fluctuated as a result, and during the nineteenth century industrial demand for orchil pushed this trade to a global scale. When a new harvesting area was found, prices normally went down but were affected by other factors, including European wars, which interfered with trade routes. Trade could be competitive, particularly between agents, and merchants hung on to stocks in expectation of higher prices.

Orchil lichen was harvested by hand and packed into bales for shipping. Before about 1830 the orchil trade sourced supplies from such places as the Canary Islands, the Azores, Madeira, and Cape Verde.[25] Within twenty years, activities had widened to Angola and other parts of Africa, as well as South America, Ceylon, Madagascar, and even further afield. The same species of lichen was normally gathered at each harvesting site but could contain quantities of twigs, loose dirt, or stones. In the Canary Islands, poorly paid gatherers would actively mix lichen with small stones or vegetable material to add weight.[26] This was sometimes cleaned before shipping, adding to the cost. The dyestuff was purchased by bale weight, and, in the case of the Lisbon agent with whom Bedford II dealt, by means of prior sampling. The agent would extract samples from marked and selected bales and consign them by ship to Leeds for rapid testing and evaluation. Handwritten notes by Bedford II on the side of sample consignment despatch lists occasionally record "dusty," "dirty," or "clean," indicating that contamination and condition of the lichen was take into account as well as dye content. A price was offered on that basis and, if agreed, the bales were shipped.

Finished orchil and cudbear were sold in casks and an efficient system of ordering and delivery was essential alongside the processing. The occasional curt letters of complaint filed in the archive indicate that the quality of their product and customer satisfaction were taken seriously.

The Bedfords were not dyers; they were manufacturers who supplied dyers and tanners. Having taken on his late father's business in the 1840s, the young Bedford II proceeded to work "with great patience and industry,"[27] as his son Bedford III records, without outside financial assistance and, it is thought, mostly alone. Given the complexities and the potential pitfalls of purchasing, processing, and marketing orchil, it was an impressive achievement that the business prospered, initially under the name "J. Bedford and Co."

[25] Sánchez-Pinto, *Brief History*, 543–51.
[26] Sánchez-Pinto, *Brief History*, 549.
[27] Bedford Family History.

At an unrecorded date around 1850, Bedford II went into business with a Mr Edward Wood. Wood's family owned land off Kirkstall Road adjoining the River Aire, about a mile west of the bridge. There were several large dyehouses in the area, each taking advantage of the good water supply and communications via the river. The newly styled partnership of "Wood & Bedford" took up new premises on Wood family land off Kirkstall Road. Wood was not a chemist; his task was to work in the office and take a share of the travelling, to visit existing clients, and to find new ones. Bedford II also enjoyed being out on the road as a beneficial alternative to the laboratory, and knew all his clients personally. Sales records from the time show a wide range of clients concentrated across northern and central England, as well as London. These included the well-known Wardle dyeworks at Leek. On the death of Bedford II, a letter of condolence from Sir Thomas Wardle to his widow describes him as "one of my oldest friends." On the death of Edward Wood in 1882, the company name of Wood & Bedford was retained.[28]

Having established his orchil- and cudbear-making business, Bedford II set up a private dye lab at his home for personal experiments. After the move to Kirkstall Road, the company's dye activities had developed and expanded, and were no longer solely concerned with orchil production. However, alongside the major advances in dye chemistry with which he kept abreast, the study of orchil and lichens continued to hold a special personal interest for Bedford II. Although the company continued to process orchil well into the synthetic age, his lifelong interest extended far beyond immediate business needs. In the 1850s he corresponded with leading botanists, such as John Lindley (1799–1865), to establish correct identifications of several dye lichen species.[29] Through correspondence with merchants and agents, he researched the stories of how new lichen sources in South America and Angola were found. He donated samples of several dye lichens to institutions such as the Pharmaceutical Society and the Horticultural Society. The company participated in exhibitions; Wood & Bedford was awarded a Bronze Medal for its exhibits of lichens and dyed samples at the Great Exhibition of 1851. Bedford II himself lectured on dye-producing lichens and was "something of an authority,"[30] according to his son, Bedford III. Indeed, the archive originally included an important botanical collection of dye-producing lichens from the Canary Islands.[31]

James Bedford III wrote that his father was

[28] There were two further name changes over the next hundred years, the first in 1900 when Wood & Bedford amalgamated with ten other local companies (mainly drysalters and chemists) to become the Yorkshire Dyeware and Chemical Company (YDCC). The company acquired several other sites, but occupied the Kirkstall Road site for an unbroken 154 years. Yorkshire Chemicals, as it became known in the twentieth century, went into receivership in 2004, and in 2008 it became a demolition site.

[29] From various letters in the Bedford Family Archive.

[30] Bedford Family History.

[31] These were investigated by mycologists at Kew from 2008 to 2011, and as a direct result are now held in the collection of Economic Botany at Kew. Begoña Aguirre-Hudson, Isabella Whitworth, and Brian Spooner, "J. M. Despréaux's Lichens from the Canary Islands and West Africa: An Account of a 19th Century Collection Found in an English Archive," *Botanical Journal of the Linnean Society* 166 (2011): 185–211.

> well versed in Chemistry according to the knowledge of Science at the time, his methods were always precise and based on the best scientific principles he could adopt; his industry was prodigious and his determination to carry out original ideas was very marked. His action was always based on the results of chemical experiment, carried out usually with his own hands. … The testing department had always his special attention and until he got his son [the writer, Bedford III, then fourteen years old] into the laboratory in the year 1870 he carried out all the testing of samples and experimenting personally.[32]

The Bedfords began to adapt to a rapidly changing market in the final decades of the nineteenth century. In 1881, the company was making only cudbear, orchil, and indigo extract. In 1887, Bedford III patented a mechanised orchil vat (which he called a "magog") to speed up production in an attempt to counter serious competition in the orchil business. These magogs continued in working production until 1939.[33] As the century progressed into the synthetic age, the company manufactured hybrid natural/synthetic dyes, such as the Patent Fustins in which fustic was combined with aniline dyes from BASF. In order to follow the continuing trend and to maintain a profitable enterprise, the Bedfords eventually manufactured synthetic-only dyes. They also made inks, soaps, varnishes, and glue, and the company was involved in producing an early form of linoleum, called Lyncrusta. Additionally, the company held patents for machinery and processes such as waterproofing.

## Charles Samuel Bedford (1865–1945) and the Perkin family

As a father who had driven an expanding business since the age of eighteen, we can imagine that Bedford II had high expectations of his sons. His elder son, James E. Bedford (Bedford III) worked with his father from the age of fourteen. A second son[34] emigrated to New Zealand. The third son was Charles Samuel Bedford.

Curiosity was bred in the Bedford family. As youths, Bedford III and the eleven-year-old Charles constructed the first telephone in Britain by studying a description of Alexander Graham Bell's new invention in *Scientific American*. They are thus credited with making the first British phone call at Sycamore Lodge, Leeds.[35] In 1911, seven years after his father's death, Charles wrote an account of his early years in the family business. He wrote with some resentment of his older brother Bedford III, claiming that it was he, Charles, who had possessed the vision in pushing the business forward by "taking up new things" in the face of

[32] Bedford Family History.

[33] Maurice Rayner Fox, *Dye Makers of Great Britain 1856–1956: A History of Companies, Products and Changes* (Manchester: Imperial Chemical Industries, 1987), 136.

[34] Henry Denison Bedford.

[35] "Secret Lives of Objects," Leeds Museums and Galleries, http://secretlivesofobjects.blogspot.co.uk/2011/06/first-uk-telephone-call.html (accessed 14 November 2013).

the growing aniline market.[36] Charles wrote that at an age when he was too young to choose for himself (he was fifteen) he was "induced" by his father to join the family works. Because of his "inclination for chemistry" he was sent to the Yorkshire College "to patronise the new dyeing school, altogether a mistake as I should have been sent away from home life abroad to have studied at a German University."[37] Charles was aware that German universities offered research facilities and laboratories simply not available in Britain.[38] During the 1880s many Britons chose to study in Germany; for instance, thirty-three studied with Adolf Baeyer in Munich.[39] One of Baeyer's outstanding achievements was elucidating the structure of the indigo molecule.[40]

Rather than going to Germany, however, Charles Samuel Bedford studied dyeing and chemistry for a year at the Yorkshire College in 1880–81 under Professor J. J. Hummel. There was one other chemistry student in that year: Arthur George Perkin, second son of William Henry Perkin. Perkin had famously discovered "aniline purple" or mauveine in 1856 and made a fortune from the dye.[41] The meeting of Charles and Arthur at the Yorkshire College was to have lifelong consequences. The two men became friends and colleagues, and in 1887 Arthur married Charles's sister Annie. It appears that Arthur and Annie had a happy marriage and Annie would often work in the laboratory with her husband. They had no children and lived in Leeds until their deaths.[42] Relations between the extended Perkin and Bedford families were cordial; a personal letter from Perkin to Bedford II advises him on treatment for his gout.[43] Records of financial investments show that the lives of the Bedford and Perkin families were closely entwined, and the Bedford collection includes some of Arthur's papers, notebooks, photographs, and notes. Charles and Arthur worked together on at least one dye project, publishing a paper on one of the colourants from old fustic (*Morus tinctoria*) in 1895.[44]

Arthur Perkin joined the staff of the Dyeing Department at Yorkshire College in 1892. His research work focused on the isolation of natural colouring matters, as synthetic dyes did not appeal to him and "he was content to leave to others the

[36] Set of uncatalogued, handwritten notes from the Bedford archive: intended recipient unknown.

[37] Uncatalogued notes, Bedford archive.

[38] Jack Morrell, "W. H. Perkin, Jr., at Manchester and Oxford: From Irwell to Isis," *Osiris* 8 (1993): 105–107.

[39] Joseph Fruton, "Contrasts in Scientific Style: Research Groups in the Chemical and Biochemical Sciences," *Memoirs of the American Philosophical Society* 191 (1990): 141–42.

[40] Adolf Baeyer, "Ueber die Verbindungen der Indigo-gruppe," *Berichte der Deutschen Chemischen Gesellschaft* 16 (1883): 2188–204; see also Carsten Reinhardt and Anthony S. Travis, *Heinrich Caro and the Creation of Modern Chemical Industry* (Dordrecht: Kluwer, 2000).

[41] Simon Garfield, *Mauve: How One Man Invented a Colour that Changed the World* (London: Faber and Faber, 2000), 43. Here Garfield draws on Anthony S. Travis, *The Rainbow Makers: The Origins of the Synthetic Dyestuffs Industry in Western Europe* (Bethlehem, PA: Lehigh University Press, 1993). Travis also deals with orchil "as a standard dyestuff throughout Europe before the synthetic age," with particular reference to a French rival to Perkin's dye known as "French purple." The French dye was claimed to act quite quickly.

[42] F. M. Rowe and R. Robinson, "Obituary Notices of Fellows of the Royal Society, Arthur George Perkin. 1861–1938," *Journal of the Chemical Society* 2 (1938): 1738–54.

[43] Sir William Henry Perkin to James Bedford II, date unknown, now with Dr Howard Varley, archiving collection for the West Yorkshire Archive Service.

[44] Charles Samuel Bedford and Arthur George Perkin, "Some Derivatives of Maclurin," *Journal of the Chemical Society, Transactions* 67 (1895): 933–36.

confirmation of his deductions."[45] He established an international reputation which attracted chemists from abroad to work at Leeds under his guidance.[46] In 1918 Arthur Perkin published *The Natural Organic Colouring Matters* with Arthur Ernest Everest, which included a chapter on the indole group.[47] In that chapter was a section titled "The Purple of the Ancients," since by then it was recognised that the famous molluscan pigment, Tyrian Purple, was an indigoid dyestuff, but there is no mention that Perkin worked on shellfish purple himself.

William Henry Perkin's eldest son was William Henry Perkin, Jr. (1860–1929). He and his brother Arthur studied together as children, undertaking experiments set for them by their father in a hut in their garden at Sudbury.[48] Eventually obtaining his father's permission to study in Germany, the younger Perkin studied for his Ph.D. from 1880 to 1882 under Johannes Wislicenus in Würzburg. He later moved to the University of Munich in 1882 as Privatdozent and research assistant to Adolf Baeyer. Baeyer remained a revered scientific and intellectual hero to Perkin for the rest of his life.[49] Perkin remained in Munich until 1886, when he returned to Britain. From 1892 he held the chair of organic chemistry at Owens College, Manchester; in 1907 he set up a school of organic chemistry that was considered a bright star in the skyscape of British chemical research.[50] Perkin and his brother Arthur published papers together on the derivatives of anthraquinone and kamala, an Indian dyestuff.[51]

Baeyer had another outstanding student besides Perkin, namely Paul Friedländer (1857–1923) whose work on shellfish dyes we will encounter below. It is quite possible that the paths of Friedländer and Perkin crossed at that time.

## The mysterious "Tyrian Purple" sample

The Leeds archive relates closely to commercial dye manufacture, whether natural, synthetic, or in combination. So it was surprising to find a small, rectangular, beige envelope fall from where it had caught in the hinge of a trunk. On the envelope, written with a beautiful flowing script in red ink, was the label "Tyrian Purple. Dyed with the bodies of shellfish found on the African Coast. C. S. Bedford" (Figure 5, left-hand side).

Inside the envelope were two twists of purple cotton strands; each twist is loosely knotted, weighs approximately 0.4 gram, and comprises about twenty-four individual strands (Figure 5, right-hand side). The sample is not dated, and no supporting

[45] Rowe and Robinson, "A. G. Perkin," 1740.
[46] Rowe and Robinson, "A. G. Perkin," 1742.
[47] Arthur George Perkin and Arthur Ernest Everest, *The Natural Organic Colouring Matters* (London: Longmans, 1918).
[48] Rowe and Robinson, "A. G. Perkin," 1738.
[49] Morrell, "W. H. Perkin, Jr.," 105.
[50] Morrell, "W. H. Perkin, Jr.," 105.
[51] Rowe and Robinson, "A. G. Perkin"; Arthur G. Perkin and W. H. Perkin, "On Some Derivatives of Anthraquinone," *Journal of the Chemical Society, Transactions* 47 (1885): 679–84; Arthur G. Perkin and William. H. Perkin Jr., "Notiz über 'Kamala'," *Berichte der Deutschen Chemischen Gesellschaft* 19 (1886): 3109–10.

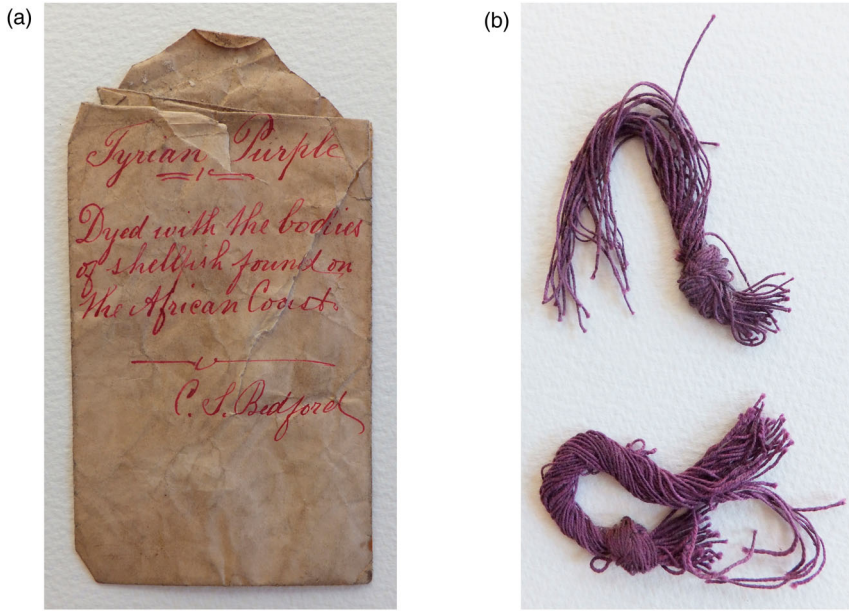


FIGURE 5 Left: envelope signed by Charles Samuel Bedford (width and height, when closed, 6.2 x 10.5 cm). Right: twists of purple-dyed cotton yarns found inside the envelope (length of each twist about 9 cm).

documentation has been found related to this item. The handwriting is definitely that of Charles Samuel Bedford.

Finding a genuine Tyrian Purple dyeing at such a relatively modern date would be a significant discovery, since the use of authentic Tyrian Purple, based on a molluscan dyestuff source, ceased about a millennium ago. Named after Tyre, the capital of ancient Phoenicia whose craftsmen developed this "dye of dyes," the fabulously expensive Tyrian Purple typically adorned the vestments of sovereigns, generals, and priests, beginning nearly four millennia ago.[52] The "lost" shellfish dyes of the ancients intrigued early modern commentators. Fabio Colonna, a member of the Accademia dei Lincei, wrote on Tyrian Purple in 1616, and in 1685 the English naturalist William Cole published his findings on purple-producing molluscs of Ireland, Wales, and north Somerset.[53] In a much published (and translated) work on ancient history, Charles Rollin (1661–1741) included details of trade in shellfish purple-dyed cloths, basing his observations on classical references, and the research of others.[54] Reaching further back in history, we note that the first known Englishman

[52] Koren, "New Chemical Insights," 43–67.

[53] Fabio Colonna, *Purpura, hoc est, De purpura ab animali testaceo fusa* … (Rome: apud Jacobum Mascardum, 1616), republished with commentary by Johann Daniel Major, *Fabii Columnae tractatum de Purpura Annotationes* (Kiel: Reumannus, 1675); William Cole, "A Letter from Mr William Cole of Bristol, to the Phil. Society of Oxford: Containing His Observations on the Purple Fish," *Philosophical Transactions* 15 (1685): 1278–86.

[54] Charles Rollin, *The Ancient History of the Egyptians, Carthaginians, Assyrians, Baylonians, Medes and Persians, Grecians and Macedonians* (Cinicinnati: Conclin, 1842), vol. II, 369–70.

to write about sea snail dyes was the Venerable Bede (672/673–735) in his *Historia Ecclesiastica Gentis Anglorum*:

> There is also a great abundance of snails, of which the scarlet dye is made, a most beautiful red, which never fades with the heat of the sun or exposure to rain, but the older it is, the more beautiful it becomes.[55]

This fascination with Tyrian Purple continued into the Bedford era, reflecting wider scientific and literary interests. For example, Tyrian Purple (which Shakespeare mentioned in his plays) appears in the poetry and literature of the nineteenth century.[56] The purple dye originally produced from snake excrement[57] was termed "murexide" by Friedrich Wöhler and Justus Liebig for its perceived similarity to the colour of the purple Tyrian dye, said to have been prepared from a species of murex, a shellfish.[58]

Perkin Sr., famous for his marketing skills, originally named his own mauve dye "Tyrian Purple" for its colour, to associate it with the status of emperors and other high ranking individuals, thereby investing it with perceived value. Perkin's mauve was, of course, not of molluscan origin. Perhaps Charles S. Bedford was just following in Perkin's footsteps when he labelled his sample: he may have received the sample from another source, and named it "Tyrian Purple" because its colour suggested ancient purple; or he may have been told that it was dyed with shellfish. On the other hand, the authentic character of the archive and the meticulous work and chemistry of the Bedfords suggested to us that the Tyrian Purple samples would probably not have been labelled in error. If the samples had not been dyed by Charles, it is likely he would have known who had done so, and trusted the information he had been given.

## Analysing the Bedford "Tyrian Purple"

In order to resolve the mystery of the Leeds "Tyrian Purple," we carried out microscopic and chromatographic analyses to ascertain whether the red ink and the purple dyeing were indeed produced from mollusca. The chromatographic analyses were performed via HPLC (High-Performance Liquid Chromatography).[59]

[55] Venerabilis Bedae, *Historia Ecclesiastica Gentis Anglorum* (London: Sumptibus Societatis, 1838), 8, https://archive.org/stream/bedehistoriaecc00bedegoog#page/n82/mode/1up (accessed 21 December 2014); A. M. Sellar, *Bede's Ecclesiastical History of England* (London: Bell, 1907), 5, https://archive.org/stream/bedesecclesiast00bedegoog#page/n53/mode/1up (accessed 21 December 2014).

[56] Lord Byron (George Gordon), *The Destruction of Sennacherib*, http://www.poetryfoundation.org/poem/173083 (accessed 16 November 2015); Christina Rossetti, *A Birthday*, http://www.poetryfoundation.org/poem/174258 (accessed 16 November 2015); L. E. Landon, *The Broken Spell*. http://www.poetrynook.com/poem/broken-spell-first-proven%C3%A7al-minstrels-lay-fairy-tale (accessed 16 November 2015); Robert Browning, *Popularity* (accessed 16 November 2015); Emily Dickinson, "Purple is in Fashion Twice" (accessed 16 November 2015); Oscar Wilde, *The New Helen, Serenade, The Garden of Eros*, http://www.bartleby.com/143/index1.html (accessed 16 November 2015).

[57] F. Wöhler and J. Liebig, "Untersuchungen über die Natur der Harnsäure," *Annalen der Chemie und Pharmacie* 26 (1838): 241–336, on 319–27; see also Travis, *Rainbow Makers*.

[58] George Fownes, *Elementary Chemistry: Theoretical and Practical* (Philadelphia, PA: Lea & Blanchard, 1845), 384.

[59] Details of this technique for the study of molluscan purple have been discussed in, e.g., Zvi C. Koren, "Archaeo-Chemical Analysis of Royal Purple on a Darius I Stone Jar," *Microchimica acta* 162 (2008): 381–92; Koren, "A New HPLC-PDA Method for the Analysis of Tyrian Purple Components," *Dyes in History and Archaeology* 21 (2008): 26–35.

The identification of a possible malacological source for the Leeds "Tyrian Purple" – as for any malacological purple – is based on the detection of certain chromatic biomarkers that are only produced by such marine species. These trademarks are brominated colourants that are based on the indigo molecule. The chemical constitutions of purple molluscan pigments have been studied in detail. The pigment, which is produced from the extracted precursors contained in the hypobranchial gland of the living animal, can consist of as many as about ten colourants.[60] However, the major dye components are usually the three indigoids – the red-purple dibromoindigo (DBI), violet monobromoindigo (MBI), and blue indigo (IND). The mollusca that can produce these colourants in the Mediterranean waters are the Muricidae family of sea snails, commonly known as *Murex* snails,[61] the modern taxonomic names for which are *Hexaplex trunculus*, *Bolinus brandaris*, and *Stramonita haemastoma*. Shellfish dyes capable of yielding purple pigments have a wider spread than the Mediterranean. For example, around the European and northwest African coasts in the North Atlantic Ocean there exists the *Nucella lapillus* species; near Australia, the *Dicathais orbita*; in Japanese waters, the *Rapana venosa* and others; and from Central America the famous Oaxaca *caracol* snails (*Plicopurpura pansa*) are used even today.

Since it was possible that, if the dyeing were genuine, Charles might have used shellfish pigment for the ink, we also analysed the beautifully penned and coloured reddish ink from the envelope along with the dye on the cotton sample. A very small fragment of paper with ink was removed from beneath an existing tear and examined. However, the analytical chromatographic results of the extraction of this ink showed that it did not contain any of the above-mentioned markers and thus was not produced from a molluscan source. The remainder of our analysis therefore focuses on the dyeing itself.

## *Microscopic analysis*

Prior to chemical analysis, the yarn samples were inspected and photographed at increasing magnification under a stereomicroscope. Photomicrographic images revealed that the samples were uniformly coloured, as can be seen from Figure 6. This homogeneity of colour indicates that the cotton yarns, if coloured with a molluscan source, were *not* coloured by means of smearing the glandular secretions onto the textile and then allowing the precursors to develop into their final purple colour when exposed to the sun and the oxygen in the air. This photochemical oxidation technique, as practised for example with the *caracol* secretions in Oaxaca, Mexico, produces unevenness of colour on cloth or yarn that can be visible to the naked eye. Thus, the microscopic analysis of the Bedford cotton yarns demonstrated that, if they were coloured with a molluscan source, they must have undergone true

[60] Koren, "Archaeo-Chemical Analysis," 381–92.
[61] Ehud Spanier, ed., *The Royal Purple and the Biblical Blue, Argaman and Tekhelet* (Jerusalem: Keter, 1987).

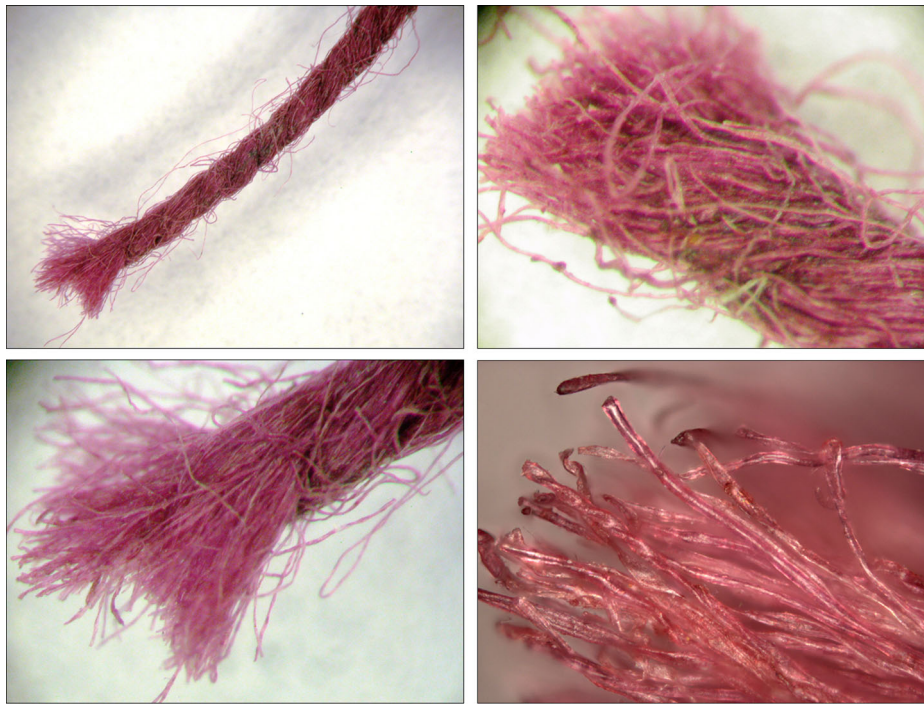

FIGURE 6 Photomicrographic pictures obtained with a stereo microscope of the fibres in the purple-dyed cotton of Fig. 5.

vat-dyeing, which required reduction and oxidation processes that would produce a uniform colouration.

### *Chromatographic analysis*

The purple dye on a few millimetres of cotton yarn was extracted and analysed via the HPLC method according to previously published procedures.[62] The resulting chromatogram and column chart (Figure 7) show that most of the purple dye was composed of red-purple dibromoindigo (DBI), with a small quantity of blue indigo (IND), and an even smaller amount of violet monobromoindigo (MBI). The presence of minor amounts of the MBI and IND dyes may be due to their natural existence in the molluscan pigment; alternatively, they may have been unintentionally but artificially produced by the photo-debromination of the mainly DBI colourant during the dyeing stage, since it probably was performed in the presence of light. The presence of brominated dyes in a pigment is conclusive evidence that this cotton dyeing comes from a molluscan source. Thus, as written on the envelope, the cotton yarn dyeings are indeed Tyrian Purple dyed with shellfish – a remarkable result for such a relatively modern date.

[62] Koren, "Archaeo-Chemical Analysis," 381–92.

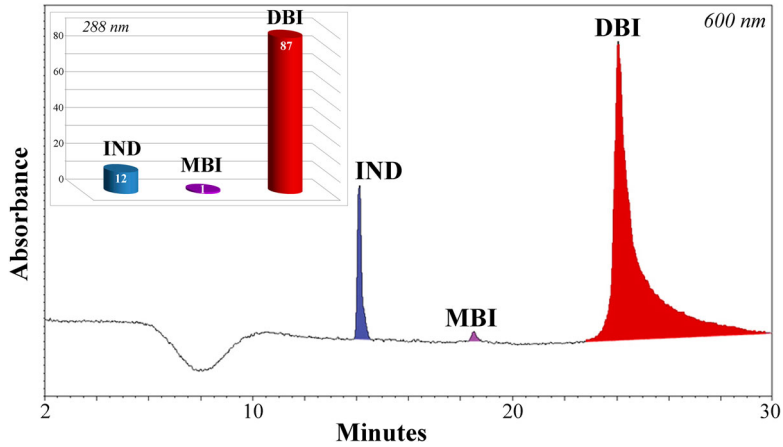


FIGURE 7 HPLC chromatogram (shown at 600 nm for visualisation purposes) of a DMSO-extract of the purple dye on the cotton yarns indicating the three detected indigoids present in the purple pigment; in the inset the relative peak areas are calculated at the standard 288 nm wavelength.

## *Malacological provenance*

We can only speculate regarding the specific molluscan species used for the Bedford dyeing. Since the resultant dye components show that there is only a trace amount of MBI in the pigment, *H. trunculus* can be ruled out. Koren has previously established that *H. trunculus* can be distinguished from all other molluscs by the fact that this species produces a significant amount of MBI in its pigment, whereas other snails generally lack this dye.[63]

Charles S. Bedford's handwritten comment on the envelope states that it was dyed with shellfish found on the "African Coast." But to which coast was he referring? Sea coasts are the source of many lichen species used in orchil, and we could speculate that the shellfish came from the same location as the orchil lichen purchased by the Bedfords. By the end of the nineteenth century, sources of Bedford lichen dyestuff included Ecuador, Galápagos, Angola, and Mozambique. Sources closer to home, such as the Canary Islands, the Azores, Madeira, and Cape Verde, had been largely exhausted much earlier in the century. The Bedfords did not have employees abroad buying lichen dyestuff; they dealt through agents in Lisbon, London, and Liverpool. Given commercial practices of the time it is highly unlikely that these agents would have been involved in obtaining shellfish material for the dyeing. It is possible that a colleague or a contact travelling abroad knew how to extract the pigment from the snails at the coast, perhaps by watching the locals doing it, and transported it back to Leeds. We could also reason that Arthur Perkin, Charles's close associate and brother-in-law, who had many international professional contacts, may have given the dyed sample to Charles.

[63] Koren, "Archaeo-Chemical Analysis," 381–92.

Based on the geographical description of "shellfish found on the African coast," the non-*trunculus* mollusca native to this area and that are known to produce a purple indigoid pigment include *B. brandaris* and *N. lapillus* (both inhabiting the North Atlantic Ocean), and *S. haemastoma* (found in both the North and South Atlantic oceans).[64] These locations encompass the whole western African coast. Of course, some other related sea snail species may have been used. Based on previous estimations,[65] in order to dye the Bedford cotton yarns to the dark purple colour we see, the pigment from about twenty or thirty such snails would have been needed, which is not a difficult number to collect.

Where might the actual dyeing have taken place? Were the cotton yarns also dyed "on the African coast" close to where the sea snails were collected, or did the dyeing take place, for example, in a laboratory? True vat dyeing, as in the case of indigo, requires a reducing agent to convert the water-insoluble molluscan purple pigment into a soluble less-coloured or "whiter" (leuco) reduced textile dye. If the Bedford dyeing was performed at the coast it is likely that the dyer would have used a natural reducing agent, probably the bacterium naturally present in the glandular flesh extracted from the snail.[66] This was the natural vat-dyeing method of antiquity, a process that required a few days to convert all of the pigment to a reduced and soluble form; experience shows that the bacteria are a weak reducing agent.[67]

However, it is unlikely that the dyeing was done in this ancient fashion, as knowledge of the ancient method was lost until recently rediscovered by Edmonds, Koren, and Boesken-Kanold.[68] Thus, dyeing at the African coast where the snails were collected is unlikely. In order to dye with shellfish pigment, a solid chemical foundation was needed, as described below. We believe that the snail pigment was used for dyeing by expert chemists.

### *Dating the Bedford Purple*

Dating the dyeing requires further detective work. Two pre-existing chemical discoveries were necessary in order to be able to uniformly dye the samples with such expertise. One was knowledge that the main pigment contained in shellfish dyes was an indigoid, which, as with indigo (as mentioned above), requires a reduction process for the dissolution of the pigment. The other was the understanding that to dissolve an indigoid requires the hydrosulphite reduction vat, which was developed by chemists Paul Schützenberger and Felix de Lalande in 1873 for the first use of vat dyeing and printing with natural indigo.[69]

[64] WoRMS (World Register of Marine Species), http://www.marinespecies.org (accessed 16 November 2015).

[65] Koren, "New Chemical Insights," 43–67.

[66] As identified by Zvi C. Koren, "The First Optimal All-Murex All-Natural Purple Dyeing in the Eastern Mediterranean in a Millennium and a Half," *Dyes in History and Archaeology* 20 (2005): 136–49; Koren, "New Chemical Insights," 43–67.

[67] Koren, "Purple Dyeing," 136–49.

[68] Koren, "Purple Dyeing," 136–49.

[69] P. Schützenberger and F. de Lalande, "Note sur un procédé nouveau de teinture et d'impression, au moyen de l'indigo," *Bulletin de la Société Chimique de France* 20 (1873): 7–11; Schützenberger and de Lalande, "Improvement in the Preparation of Indigo for Dyeing and Printing," *US Patent* No. 136,770 (1873). The strong reducing agent is

As for determining the chemical constitution of shellfish purple, important research was carried out by Edward Schunck in Manchester. Schunck published analyses of shellfish pigment in 1879 using sea snails gathered in Britain, and in 1880 using a sample coloured in Nicaragua.[70] He noted that the substance was somewhat similar to indigo, but he did not definitively determine the molecular constitution of the pigment.[71]

Paul Friedländer's work at the Technologisches Gewerbemuseum in Vienna in 1909 determined that the main component of the purple pigment from the shellfish which he analysed was an indigoid, namely dibromoindigo.[72] This discovery provides the limiting date for establishing the approximate period for the dyeing of the Bedford Tyrian Purple: the cotton yarns cannot have been dyed before 1909.

## *The identity of the purple dyer*

We may never know the identity of the dyer of this Tyrian Purple sample. In addition to the Bedfords themselves, it may have been Friedländer or one of the Perkin children, or perhaps the leading dye chemist Arthur George Green who came to the Yorkshire College (later the University of Leeds) as chair of Colour Chemistry and Dyeing from 1903.[73] However, the family, scientific, and working connections of the Bedfords offer some context for understanding the sample's retention in the archive. The Bedfords keenly studied recent chemical research, and any new publication would soon have reached them. Bedford II had a wide circle of correspondents at a high scientific level;[74] he was a member of the Leeds Philosophical and Literary Society where regular lectures were held. His chemical expertise was passed to his children Bedford III and Charles Samuel Bedford over the course of a lifetime, before he passed away in 1903.

If Bedford III was the dyer (or if he received the sample from another source), it must have been dyed after 1909, when, as stated above, Friedländer discovered the indigoid nature of the pigment, but prior to 1914, when Bedford III became Lord Mayor of Leeds. The probable timeframe in which he might have dyed it, or had it dyed, is thus *ca.* 1910–13. However, during this time he was executive director of the company and probably not much taken up with chemical studies and experiments. Additionally, in 1911, Charles S. Bedford describes Bedford III being

[69] *Continued*
sodium hydrosulphite, also referred to as sodium dithionite, $Na_2S_2O_4$, and requires only a few minutes to reduce indigo.

[70] Edward Schunck, "Notes on the Purple of the Ancients," *Journal of the Chemical Society, Transactions* 35 (1879): 589–96, and 37 (1880): 613–17.

[71] Contrary to what other authors have written on the subject, all of Schunck's samples were "coloured" with shellfish purple and *not* dyed. Colouration was accomplished by the technique of smearing the glandular secretions of the snail onto a textile and developed under the action of sunlight, as previously mentioned.

[72] P. Friedländer, "Über den Farbstoff des antiken Purpurs aus murex brandaris," *Berichte der Deutschen Chemischen Gesellschaft* 42 (1909): 765–70. However, there are no known dyed samples from his research.

[73] We thank an anonymous reviewer of this paper for this information.

[74] From the Bedford archive: letters from John Lindley (1850 and 1851), which include references to opinions of Dr Jonathan Pereira, Sir William Hooker, and a "Mr Bennett" of the British Museum – probably John Joseph Bennett; from Asiatic Society (1853) relating to a request about the cuneiform research of Sir Henry Rawlinson.

hampered by the "distraction of other duties" and betrays a lack of accord, to the point of resentment, between himself (Charles) and James.[75]

If one of the Bedfords was the dyer, a more likely possibility is that Charles himself was responsible for the dyeing, perhaps assisted by a close associate or colleague. His participation in the adventure could very well explain why no other notes have been found relating to the Tyrian Purple dyeing. A thorough understanding of state-of-the-art chemistry of that time was essential, which Charles possessed, participating as he did in a prominent scientific circle of those "in the know" regarding the chemical nature of molluscan pigment. Charles Samuel Bedford and Arthur Perkin were friends and working colleagues, and the marriage of Annie Bedford to Perkin cemented scientific and social links between two strongly scientific families. The latest chemical knowledge needed for the molluscan purple dyeing was available to Charles Bedford through this wider group, which may also have included Paul Friedländer through the Munich connection with W. H. Perkin, Jr.

Tyrian Purple continued to be studied by nineteenth-century scientists whose research would have been familiar to the educated Bedford family. In addition to the research of Edward Schunck (who described the dye as "The Purple of the Ancients"), Henri de Lacaze-Duthiers (1821–1901), botanist and anatomist, studied purple-producing *Muricids* from the coasts of the Mediterranean and Atlantic, from the late 1850s.[76] He devised a photographic process using purple pigment obtained from "Purpura Lapillus" (*Nucella Lapillus*) and reproduced images on textiles and paper. Lacaze-Duthiers also made drawings using the snail secretions.[77] The reconstruction of the ancient dye which found its way into the Bedford archive took place in a context of intense interest in obtaining purple pigments.

## Conclusions

The Bedford archive is valuable for both its textual and material sources associated with a family of industrial chemists from Leeds. Trade records and letters in the archive give insights into communication, exploration, botanical naming, and chemical advances in the lichen trade, and reveal the pressure on natural lichen stocks affected by heavy trade demand.[78] Successive family members were not only concerned with business profits, but also displayed interests in culture, education, and the arts,[79] in addition to advancing chemical knowledge of their main

[75] Set of handwritten notes from the archive, as at n. 36: intended recipient unknown.

[76] Henri de Lacaze Duthiers, "Mémoire sur la Pourpre," *Annales des Sciences Naturelles, Zoologie*, 4th ser., 12 (1859): 5–84; "Natural History of the Purple of the Ancients," *Proceedings of the Royal Society of London* 10 (1859–60): 579–84.

[77] Dominique Cardon, Inge Boesken Kanold, Takako Terada, "Colour Purple from the Atlantic to Faraway Shores," (exhibition catalogue [s.n.], *Aquarium of La Rochelle and ISEND*, 2011).

[78] Isabella M. S. Whitworth, "An Unsustainable Demand: The Trade in Orchil Lichens Seen Through a 19th Century Archive," *Text Journal* 38 (2010–11): 4–9; Begoña Aguirre-Hudson et al., "J. M. Despréaux's Lichens," 185–211.

[79] For example, Francis Bedford (1866–1904), younger brother of Bedford III, was a distinguished architect; with Sidney Kitson he designed Leeds School of Art. Bedford II was involved in setting up the Yorkshire Institute and Village Library, and a member of the Leeds Philosophical and Literary Society.

commodity, orchil. The Bedfords and the related Perkins family were involved in experimenting, replicating, and investigating chemical substances, leaving behind intriguing material evidence of their dyeing activities which offers a valuable supplement to the other items in the archive.

Using modern chromatographic analytical techniques, we have confirmed that the sample of Tyrian Purple dyeing was a genuine example of shellfish dyeing, the most modern historic example of genuine Tyrian Purple yet found. Unfortunately, Bedford III left no explanation in the Family History for the presence of this sample, and we may never know when, why, or by whom it was dyed. At first glance, the shellfish-dyed artifacts appear out of context in the collection, representing a historical dye with no commercial potential, long fallen from use. Molluscan purple had no obvious direct connection to the lichen trade or other products manufactured by the Bedfords. However well-known as a literary subject, dyeing with shellfish was not a commercially viable process in the nineteenth or twentieth centuries.

The family may have retained the sample for its scientific interest, and also to reflect the wider nineteenth century interest in "the purple of the ancients." The colour itself doubtless helped to inspire the Bedfords' curiosity, given that their family enterprise was founded on producing another purple dye. The Bedfords' commerce in orchil purple produced from lichens – exotic plant sources from far away coastal regions – may have contributed to Charles Bedford's interest in another exotic purple produced from mollusca. Whatever the reason, the survival of these purple threads offers deeper insight into the family's engagement with materials and practices than can be gained from their textual archive alone.[80] The result is that we can identify three purples within this Bedford family history: Perkin's mauve, lichen purple, and now the "lost" Tyrian Purple, appearing like the proverbial "phoenix arising from the ashes" – or, more accurately, like a "Phoenician arising from the rubble" of the Bedfords' old Kirkstall Road site.

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[80] For recent work on material culture in the history of science, see, e.g., Pamela H. Smith, Amy R. W. Meyers, and Harold J. Cook, eds., *Ways of Making and Knowing: the Material Culture of Empirical Knowledge* (Ann Arbor, MI: University of Michigan Press, 2014); Ursula Klein and Wolfgang Lefèvre, *Materials in Eighteenth-Century Science: A Historical Ontology* (Cambridge, MA: MIT Press, 2007).

colourants. We would also like to acknowledge our appreciation to all the reviewers who offered helpful guidance for this paper.

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